

NMRMetaboWizard

Supplementary Software Documentation and User Guide

Version 0.1.0 release-candidate documentation

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Repository placeholder: https://github.com/YOUR_GITHUB_USERNAME/NMRMetaboWizard

Documentation placeholder: <https://nmrmetabowizard.readthedocs.io>

This document is intended as journal supplementary information and as a printable companion to the online user documentation. Replace all URL, DOI, ORCID, version, and licensing placeholders before submission.

1. Software overview

NMRMetaboWizard is an interactive Shiny for Python application for the stepwise preprocessing and analysis of one-dimensional Bruker 1H NMR metabolomics data. The software integrates raw FID handling, spectral preprocessing, clinical metadata alignment, exploratory analysis, outlier screening, and machine-learning evaluation in one auditable interface.

The application is intended for research and education. It is not a validated clinical diagnostic system or medical device. Biological and clinical conclusions require appropriate validation, quality control, statistical expertise, and domain review.

1.1 Core design principles

- Stepwise inspection of raw and processed signals.
- Optional processing steps can be applied, adjusted, or skipped.
- Spectral features, class labels, and clinical metadata remain conceptually separate.
- Interactive plots support zooming, panning, and export of the plotted data.
- Outliers are flagged for review rather than removed automatically.
- Imputation, scaling, and optional PCA are fitted inside the ML pipeline.
- Processing parameters and ML runs can be downloaded for reproducibility.

1.2 Software architecture

The user interface and reactive workflow are implemented in `app.py`. Scientific NMR-processing functions are implemented in `nmr_pipeline.py`, whereas metadata alignment, EDA, outlier detection, and machine learning are implemented in `clinical_analysis.py`. This separation enables testing of scientific functions independently of the user interface.

2. Availability, license, and versioning

Before manuscript submission, the source code should be published in a public GitHub repository or made available through confidential reviewer access. A tagged release corresponding exactly to the manuscript should be archived with Zenodo to obtain a permanent DOI.

- Proposed repository: https://github.com/YOUR_GITHUB_USERNAME/NMRMetaboWizard
- Proposed documentation: <https://nmrmatabowizard.readthedocs.io>
- Release tag: v0.1.0
- Archived DOI: replace with the final Zenodo DOI
- Proposed license: MIT, subject to institutional confirmation

The repository includes CITATION.cff, CITATION.bib, a changelog, contribution guidance, security and privacy guidance, automated tests, continuous-integration workflows, and Read the Docs configuration.

3. Installation and execution

3.1 System requirements

- Python 3.10 or newer.
- Windows, macOS, or Linux.
- A modern browser; Chrome or Microsoft Edge is recommended.
- Sufficient memory for the number and resolution of uploaded spectra.

3.2 Windows installation

```
cd /d C:\Users\YOUR_USERNAME\Documents

python -m venv NMRMetaboWizard\.venv

NMRMetaboWizard\.venv\Scripts\activate.bat

cd NMRMetaboWizard

python -m pip install --upgrade pip

python -m pip install -r requirements.txt
```

3.3 Starting the application

```
python -m shiny run --reload --launch-browser app.py
```

3.4 Docker execution

```
docker compose up --build
```

4. Input-data requirements

4.1 Bruker NMR archive

The uploaded ZIP archive is scanned recursively for directories containing both fid and acqu. Acquisition parameters are read from acqu and used to decode the FID byte order and numeric data type. Alternating real and imaginary values are combined to form the complex FID.

For common Bruker structures in which the experiment directory is numeric, the parent folder is treated as the biological sample identifier. Folder names therefore become study_id values and must match the clinical table.

Recommended structure:

```
study_id_001/
  1/
    fid
    acqu
```

Diagnostic or dummy acquisition directories, commonly named 99999, should be removed before creating the ZIP because they can resemble ordinary experiments but do not represent biological samples.

4.2 Clinical metadata

Clinical data can be uploaded as CSV, TSV, TXT, XLSX, or XLS. The table must contain study_id and Class columns unless alternative column names are explicitly entered in the app. Additional numeric or categorical variables may be included for visualization, correlation, stratification, or prediction.

- study_id: identifier matching the NMR folder/sample name.
- Class: cohort or outcome label.
- Optional examples: age, PSA, creatinine, prostate volume, height, weight, or Gleason score.

Only coded, de-identified data should be used in a public deployment. Variables that are consequences of the target outcome should not be used as predictors when doing so would introduce information leakage.

5. Stepwise preprocessing

All processing calculations use the full arrays. Display downsampling is used only to keep browser plots responsive and preserves local minima and maxima to reduce visual loss of sharp peaks.

Step	Default	Implementation note
Group delay	-1	Use GRPDLY; manual override or skip available
Solvent residual suppression	Enabled; $\lambda = 1 \times 10^6$	FID-domain penalized smoothing
Apodization	Exponential; LB = 1.0	Gaussian is available
Zero filling	32768 additional points	Optional
Fourier transformation	Required	FFT and FFT-shift
Phase correction	Automatic zero-order	Manual or skip available
Referencing	0 ppm; search -0.2 to 0.2 ppm	Optional
Baseline	ALS; $\lambda = 1 \times 10^6$; $p = 0.01$	arPLS and airPLS available
Alignment	Disabled	Global integer cross-correlation shift
Negative-value zeroing	Enabled	Optional through checkbox
Window	0.2–10 ppm	Can retain full range
Region removal	4.5–6.1 ppm; zero	Urine-focused default
Binning	0.01 ppm; trapezoidal	Width/count and rectangular options
Normalization	PQN	Total area, SNV, or none

5.1 Group delay and solvent residuals

Group-delay correction removes the initial complex points specified by Bruker GRPDLY or a user override. The optional solvent-residual procedure estimates a smooth real and imaginary FID component using a Whittaker-type penalized smoother with second-order differences and subtracts it from the FID.

5.2 Apodization, zero filling, and Fourier transformation

Exponential or Gaussian windows are applied to the complex FID. Zero filling appends complex zeros before the FFT. Fourier transformation is followed by FFT-shift reordering, and the ppm axis is calculated from SW_h, O1, and SFO1.

5.3 Phase, referencing, baseline, and alignment

Automatic phasing estimates a separate zero-order phase angle for each spectrum. Referencing shifts the ppm axis according to the strongest peak in the selected interval. Baseline methods include ALS, arPLS, and airPLS. The alignment implementation is a global integer shift selected by cross-correlation in one ppm window; it is not a full icoshift implementation.

5.4 Region removal, binning, and normalization

After windowing and optional negative-value zeroing, user-selected regions can be set to zero or interpolated. Binning integrates each interval using the trapezoidal or rectangular method. PQN, total-area, SNV, or no normalization can then be applied to the sample-by-bin matrix.

6. Clinical metadata alignment

The aligned analysis object contains the NMR matrix X , the class labels y , and the clinical metadata in the same sample order. Clinical variables do not automatically become predictors. The user explicitly selects NMR-only, clinical-only, NMR plus clinical, or PSA-only modelling.

The app reports matched and unmatched identifiers, duplicate clinical IDs, the number of samples with non-empty class labels, and the resulting feature count. Users should resolve unexpected mismatches before EDA.

7. Exploratory analysis

7.1 PCA and PCA loadings

PCA is fitted after median imputation and standardization of the NMR bins. The app displays selectable 2D and 3D score combinations, explained variance, and loading bar plots for one or more PCs. Loadings identify contributing bins but do not provide metabolite identification.

7.2 PLS-DA

PLS-DA uses one-hot encoded class labels and scikit-learn PLSRegression. Its score plots are intended for exploratory visualization. The displayed approximate X-variance values are not directly equivalent to PCA explained variance.

7.3 Clinical and univariate analyses

The score-color menu is populated from aligned clinical metadata. Numeric variables use continuous scales and categorical variables use discrete traces. The app also reports class counts, Welch t-tests or one-way ANOVA, clinical Spearman correlations, and clinical-bin Spearman correlations. Raw p-values should be corrected for multiple testing before confirmatory use.

8. Outlier screening

Possible outliers can be screened using Hotelling's T^2 or a robust PCA distance. Limits can be calculated globally or separately within each Class cohort. Groupwise screening is useful when cohorts are biologically shifted, because a global center may otherwise label a valid cohort as unusual.

1. Run initial EDA.
2. Detect possible outliers.
3. Inspect cohort-specific plots and the outlier table.
4. Review spectral quality, metadata, and biological plausibility.
5. Remove flagged samples, enter selected IDs manually, or skip removal.
6. Rerun EDA and compare the results.

An outlier flag is not evidence that a sample is invalid. Removal decisions should be reported and justified.

9. Machine learning

9.1 Predictors and classifiers

- Predictor sets: NMR only, clinical only, NMR + clinical, or PSA only.
- Classifiers: logistic regression, random forest, LinearSVC, and ANN/MLP.
- Default test fraction: 0.25.
- Default cross-validation: five-fold stratified CV; can be disabled.
- Optional PCA within the model pipeline: disabled by default.

9.2 Leakage control and reported metrics

Median imputation, standardization, and optional PCA are fitted inside the scikit-learn Pipeline. The app reports train and test accuracy, balanced accuracy, classification reports, confusion matrices, test predictions, cross-validation summaries when enabled, and ROC-AUC when probability outputs are available.

The current linear SVM uses LinearSVC and does not provide calibrated probabilities. Consequently, probability and ROC outputs are not produced for this classifier by the current implementation.

9.3 Feature importance

- Logistic regression and linear SVM: mean absolute model coefficient.
- Random forest: impurity-based feature importance.
- ANN: mean absolute first-layer weight, used as an approximate heuristic.

When PCA is enabled, feature importance is produced from a separate model fitted without PCA to retain original variable labels. It should be treated as exploratory rather than as a held-out estimate.

10. Output files

Module	Downloadable outputs
Preprocessing	Current plot data CSV; binned matrix; normalized matrix; processing log
Clinical alignment	Aligned sample table and mismatch summary
EDA	Current plot data, PCA variance/loadings, PLS-DA table, tests and correlations
Outliers	Outlier table and reviewed retained/removed sample set
Machine learning	ML plot data, predictions, feature importance, model-performance history

11. Reproducibility and validation

The repository includes a synthetic smoke test that exercises the NMR processing functions, metadata alignment, PCA, PLS-DA, outlier screening, and all four classifiers. GitHub Actions compile the source code, execute the smoke test, and build the documentation.

Automated software tests do not establish biological validity. Journal submission should additionally include comparison with a trusted workflow, representative spectral quality-control plots, parameter sensitivity, explicit leakage controls, and independent or nested model validation.

12. Known limitations

- Only Bruker fid/acqus input is directly supported.
- Dummy folders such as 99999 are not automatically excluded.
- Automatic phasing is zero-order only.
- Alignment is a simple global integer shift, not local warping or full icoshift.
- PLS-DA permutation testing is not currently implemented.
- Univariate p-values are not automatically FDR-adjusted.
- Linear SVM probability calibration is not implemented.
- ANN importance is heuristic.
- Cross-validation is not nested hyperparameter optimization.
- The app does not perform metabolite identification.
- The app is not validated for clinical decision-making.
- Session history must be downloaded before the app is restarted.

13. Privacy and data governance

The public repository must not contain identifiable patient data, real clinical spreadsheets, private raw FIDs, access credentials, or restricted results. A robust .gitignore is included, but git status and the complete repository history must be reviewed before public release.

The current app extracts uploaded ZIP files to an operating-system temporary directory. Secure deployments should configure access control, cleanup, retention, logging, and backup according to institutional requirements.

14. Manuscript-ready statements

14.1 Availability and implementation

NMRMetaboWizard is implemented in Python using Shiny for Python, Plotly, NumPy, pandas, SciPy, scikit-learn, Matplotlib, openpyxl, and shinywidgets. Source code is available under the MIT License at https://github.com/YOUR_GITHUB_USERNAME/NMRMetaboWizard. User documentation is available at <https://nmrmetabowizard.readthedocs.io>. The version used in this study is archived at https://doi.org/10.5281/zenodo.REPLACE_WITH_RECORD. The software requires Python 3.10 or newer.

14.2 Data availability

The source code and synthetic metadata template are publicly available in the repository. Raw clinical NMR data are subject to participant consent, ethics approvals, institutional governance, and applicable privacy regulations. Replace this sentence with the final public, controlled-access, or reasonable-request data-sharing arrangement.

15. Pre-submission checklist

- ☐ All repository and documentation URL placeholders replaced.
- ☐ Author ORCID identifiers added.
- ☐ Institutional license approval obtained.
- ☐ Repository history checked for confidential data.
- ☐ Clean installation tested.
- ☐ Smoke test and documentation workflows pass.
- ☐ A tagged release is created.
- ☐ The release is archived with Zenodo.
- ☐ The exact version and DOI are included in the manuscript.
- ☐ Example or public test data are available.
- ☐ All scientific defaults and limitations are documented.